

Retail Sales Forecasting — Final Report

1. Executive Summary

Business problem:

A retail chain operating 45 stores and 81 departments needed a weekly sales forecasting framework to support demand planning and inventory allocation, in a business context shaped by strong seasonality, holiday spikes, promotional markdowns, and significant store-level performance variation.

Approach:

An end-to-end pipeline was built covering data understanding, EDA, feature engineering, and the evaluation of five forecasting models across two families — statistical (ARIMA, SARIMA) and machine learning (Linear Regression, Decision Tree, Random Forest, XGBoost). All train/test splits were strictly chronological to prevent data leakage; XGBoost was tuned using RandomizedSearchCV with TimeSeriesSplit cross-validation.

Best performing model:

Random Forest, achieving the lowest error across every metric evaluated (MAE \$1,226.53, RMSE \$2,496.83, %RMSE 15.68%, R^2 0.987), outperforming a properly tuned XGBoost — attributable to the dataset's structure being dominated by one highly predictive feature (lag_52), which independent, deep-tree ensembling captures efficiently.

Key insights:

Seasonality (not macroeconomic conditions) is the dominant driver of sales variation; the dataset's official holiday flag mistimes the true Christmas demand peak by one week; store size/type meaningfully predicts sales; markdown impact is confounded with an underlying time trend and should not yet be treated as a proven causal driver.

Recommendations:

Build demand planning around year-over-year seasonality first; shift holiday-readiness timing one week earlier than the official flags; use Random Forest for store/department-level allocation and SARIMA for company-wide budget planning; validate markdown ROI with a controlled comparison before scaling spend.

2. Approach & Methodology

- **Data Understanding** — 420,212 rows, 45 stores, 81 departments, weekly grain, 2010–2012. No missing values or duplicates found. Data is panel data — 3,323 distinct store-department time series stacked together, not a single sequence.
- **EDA** — decomposition confirmed strong annual seasonality riding on a mild underlying trend; holiday-week sales average 7.2% higher than non-holiday weeks; store Type/Size are strong drivers; macro indicators show weak linear correlation.
- **Feature Engineering** — calendar features (week, month, quarter, year), lag features (1, 2, 4, 13, 26, 52 weeks), rolling mean/std (4-week trailing, leak-safe via shift-then-roll), and a corrected is_peak_week flag based on the EDA finding that the official Christmas flag mistimes the real peak.
- **Feature Selection** — correlation analysis showed no problematic multicollinearity; the full engineered feature set was retained, since tree-based models are inherently robust to correlated predictors.

- **Modeling** — statistical baselines (ARIMA, SARIMA) on the aggregated company-wide series; ML models (Linear Regression, Decision Tree, Random Forest, XGBoost) on the full store-department panel. XGBoost was systematically tuned via RandomizedSearchCV over 30 candidate combinations with TimeSeriesSplit(n_splits=3).
- **Evaluation** — MAE, RMSE, %RMSE, MAPE, and WMAE (5x holiday weighting, matching the original Walmart competition standard, plus an extended version also weighting the corrected true-peak week).

3. Model Comparison

Model	MAE	RMSE	%RMSE	WMAE	R ²
Random Forest	1,226.53	2,496.83	15.68%	1,281.71	0.987
XGBoost (Tuned)	1,339.64	2,810.39	17.65%	1,375.25	0.984
Decision Tree	1,489.07	3,179.38	19.97%	1,524.73	0.979
Linear Regression	1,613.06	3,007.43	18.89%	1,644.71	0.981

Table 1: ML model comparison on the held-out test set (store-department panel level).

Model	MAE	RMSE	%RMSE	MAPE
SARIMA (0,0,1)(0,1,1,52)	1,254,363.66	1,570,347.85	3.36%	2.67%
ARIMA (2,0,2)	1,528,077.22	1,952,054.85	4.18%	3.30%

Table 2: Statistical model comparison on the aggregated company-wide weekly series (not directly comparable in scale to Table 1's panel-level metrics).

Why Random Forest won: feature importance analysis showed lag_52 (same week, one year ago) dominates predictive power across all tree-based models — Decision Tree concentrates 81% of its importance there alone, while Random Forest spreads importance more evenly across lag_52, lag_1, rolling_mean_4, lag_2, and lag_26. When one feature this strongly dominates, independent deep trees (Random Forest) tend to capture the signal at least as efficiently as incremental shallow-tree boosting (XGBoost), which is the pattern observed here even after systematic tuning.

4. Business Insights & Recommendations

Which factors impact sales most?

lag_52 and other engineered lag/rolling features far outweigh macroeconomic factors (CPI, unemployment, fuel price) in predictive importance. **Demand planning should be built around year-over-year seasonality first**, with short-term trend as a secondary adjustment.

How do holidays affect sales?

Holiday weeks average **7.2% higher sales** (\$17,094 vs \$15,953). Critically, the week **flagged** as “Christmas” in the dataset is *not* the true peak — the week before it (week 51) generates roughly **2x** the sales of the flagged week itself, consistently across both observed years. Any process keyed off the raw Christmas flag will be a full week late for the actual demand spike.

Which stores/departments perform best?

Store Size correlates positively with Weekly_Sales; department- and store-level rankings show consistent, sizable spread — supporting size-tiered inventory allocation and department-level floor space decisions.

How do markdowns impact sales?

Markdown weeks show a **4.8% sales lift** within 2011–2012, but markdown activity is concentrated in years that also show an overall upward sales trend — part of this lift may be a time trend rather than the promotion itself. A controlled before/after comparison is recommended before scaling markdown spend on this basis.

Recommendations

- Prioritize year-over-year seasonal modeling over macroeconomic features for short-term forecasts.
- Shift holiday-readiness timing **one week earlier** than the raw holiday flags.
- Use **Random Forest** for store/department-level allocation and **SARIMA** for company-wide budget planning — matching model granularity to the decision.
- Treat markdown ROI claims cautiously pending a cleaner causal analysis.

5. Limitations & Future Work

- SARIMA was fit on only ~2.3 years of history (barely 2 full seasonal cycles), producing some numerical instability in coefficient estimation (flagged via covariance matrix warnings) — a longer history would improve confidence.
- Markdown effect estimates are confounded with time and require a more rigorous causal design.
- A simple Random Forest + XGBoost averaging ensemble was considered but not required, given Random Forest's clear, consistent lead across all metrics.
- Future work: per-store/department SARIMA (rather than company-aggregate only), exogenous regressors (holiday/markdown flags) in the statistical model, and monitoring for model drift in production.

6. Conclusion

An end-to-end retail sales forecasting pipeline was successfully developed and evaluated across five models spanning two methodological families. Random Forest achieved the best predictive performance by effectively capturing nonlinear relationships between historical sales, holidays, markdowns, and store characteristics. The resulting forecasting framework, along with the seasonal-timing and markdown-confound findings, can directly support demand planning, inventory management, and promotional strategy decisions.